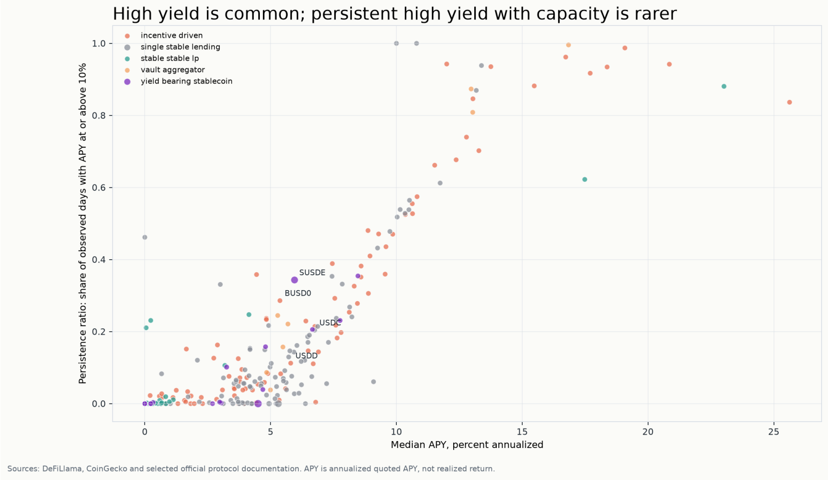


Stablecoin yield joint screen

250
pools

137,095
pool-days

2.0
median
high-yield days

The price of yield

Persistence, mechanisms and TVL response in stablecoin DeFi

A visual analytics report on why quoted stablecoin APY should be read as a regime with duration, capacity, mechanism and peg context, not as a ranked list of pools.

Scope boundary
Educational analysis only. APY is quoted annualized APY, TVL is an observed proxy, and no output is financial advice or a pool recommendation.

Research questions

RQ	Question	Evidence
RQ1	How long do high-yield regimes last?	Median APY >= 10 episode duration: 2.0 days.
RQ2	Which mechanisms explain headline APY?	Base APY coverage: 90.3%; reward APY coverage: 61.6%.
RQ3	How stable are top-yield sets?	Churn rises from 13.5% to 33.3%.
RQ4	How does peg stress change APY interpretation?	Nominal APY is harder to read when the stablecoin denominator moves.
RQ5	How can pools be compared without ranking?	38 joint-screen candidates show trade-offs, not recommendations.

Reading order
The report moves from measurement to interpretation: duration first, mechanism second, stability and context after that.

Assessment logic
A strong answer is not a top-pool list. It is a transparent comparison that keeps uncertainty visible.

Design decision
The report answers comparison questions without creating a leaderboard. Each research question adds context that a single APY column would hide.

Data sources and feasibility

Source	Role	Evidence
DeFiLlama yields	Pool universe, APY components, TVL and history	15,669 live pools observed; 250 selected for full analysis
DeFiLlama stablecoins	Stablecoin metadata and peg-price context	404 assets and 2,012 price date records verified
CoinGecko Demo	Fallback checks for selected stablecoins	Keyless sample returned 3 assets on 2026-07-08
Protocol documentation	Mechanism labels and caveats	Classification is metadata, not a safety rating

Endpoint	Observed	Minimum evidence
Yields /pools	15,669 pools	APY, APY components, TVL, chain, project and pool ID verified.
Yields /chart	503 observations in sample	Historical APY and TVL are available at pool level.
Stablecoins	404 assets	Peg metadata and stablecoin price context available.

Verified source constraint
DeFiLlama is the primary source. CoinGecko is only fallback/enrichment because public historical depth can be plan-limited.

Raw preservation
Live responses are stored as request envelopes with checksums before transformation into canonical tables.

Pipeline and canonical schema

Step	Stage	Output
1	Collect	Pool, chart, stablecoin and fallback payloads.
2	Normalize	Timestamps, APY components, TVL and identifiers.
3	Resolve	Pool IDs, stablecoin exposure, protocols and pool types.
4	Validate	Critical, error and warning checks.
5	Analyze	Episodes, survival, churn, event studies and joint screening.
6	Render	Figures, PDF, presentation previews and manifest.

Artifact	Rows	Role
pools	250	Canonical entities
pool snapshots	137,095	Clean observations
stablecoins	394	Stablecoin metadata
stablecoin prices	176,176	Peg context
risk events	7,763	Event context
pool-day panel	137,095	Analysis grain
yield episodes	9,490	Episode analysis

Entity resolution
Pool IDs preserve the DeFiLlama source identifier; stablecoin exposure uses exact IDs and documented ticker overrides.

Canonical grain
The pool-day panel is rebuilt from processed tables, not from manual spreadsheet state.

Analytical grain
The central unit is a DeFi pool observed on one UTC day. This grain supports duration, ranking churn and event-time analysis.

Quality gates and metric definitions

12
checks

0
critical failures

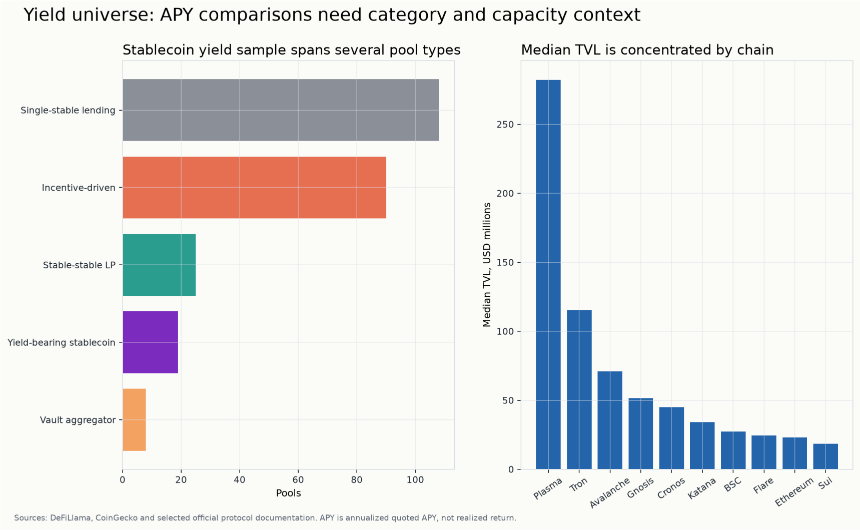
2
warnings

Check	Status	Failed
panel.exists	passed	0
pool_day.unique_key	passed	0
pool_day.apy_total.completeness	passed	0
pool_day.tvl_usd.completeness	warning	26
pool_day.chain.completeness	passed	0
pool_day.protocol_id.completeness	passed	0
pool_day.pool_type.completeness	passed	0
apy.non_negative	passed	0

Metric	Definition
High-yield episode	Continuous APY >= 10 percent run.
Survival	Kaplan-Meier with censored active episodes retained.
Ranking churn	1 minus top-k retention across horizons.
TVL response	Median normalized TVL index around APY jumps.
Frontier	Above median APY, persistence and TVL.

Quality decision
Extreme APY rows are flagged, not hidden. Removing them would make the source-reported distribution look cleaner than it is.

What is the observed stablecoin yield universe?



What is the observed stablecoin yield universe? The sample covers multiple pool types and chains, so APY should not be compared without stratification. Sample: pools=250, pool-days=137095.

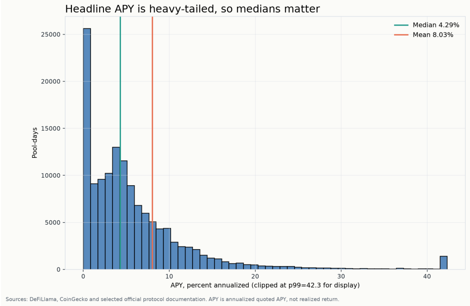
The sample spans multiple chains, protocols and pool types. A single APY number is therefore not a comparable market price.

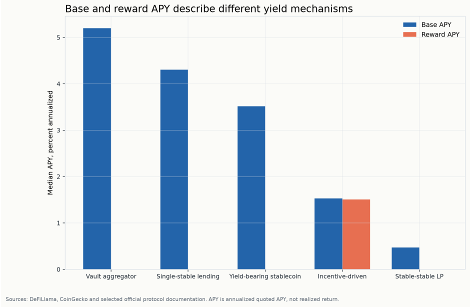
The final scope contains **250 pools** and **137,095 pool-days.**

Metric	Value
Pools	250
Chains	25
Protocols	84

Key reading
APY must be read with category and capacity context.

What creates headline APY?





Distribution and APY composition show why a single mean APY is a weak summary.

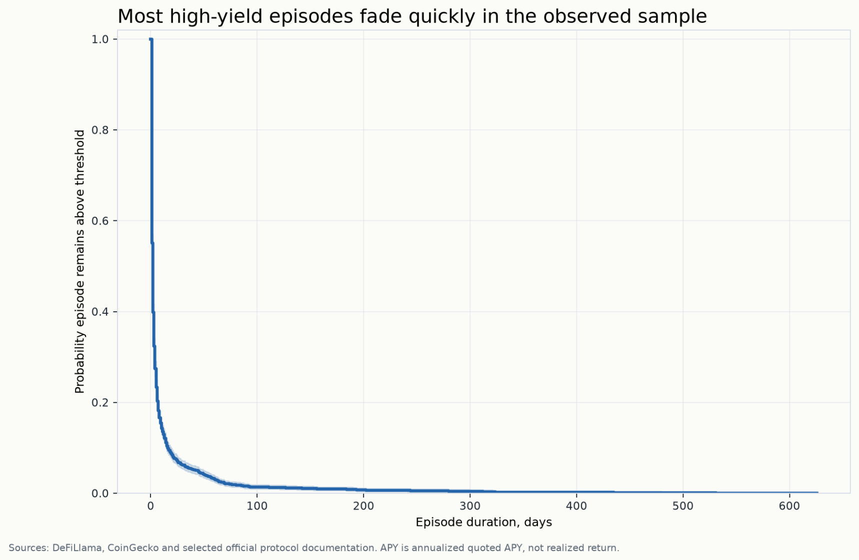
Pool-day median APY is **4.29%**, while pool-day mean APY is **8.03%**.

The gap indicates a heavy-tailed distribution: a small number of extreme observations pulls the mean above the median.

Base and reward APY describe different mechanisms. Reward-heavy pools should not be compared blindly with base-yield pools.

What it does not prove
It does not convert quoted APY into realized dollar return after fees, slippage or reward liquidation.

How long does high APY last?



How long do APY >= 10 percent episodes last? Kaplan-Meier survival declines steeply at short durations, with censored active episodes retained. Sample: episodes=2668.

The APY >= 10 percent definition produces **2,668 episodes.**

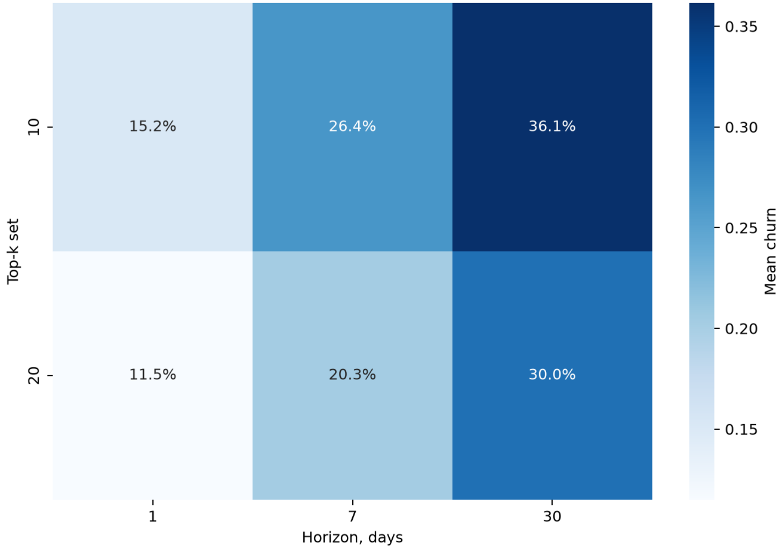
The median episode lasts **2.0 days**; by day 30, survival is only 6.2%.

Metric	Value
Median	2.0d
P75	5.0d
P90	16.0d

Key reading
High APY is usually an episode, not a durable pool trait.

Are top-yield lists stable?

Top-yield membership changes as the horizon widens



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

How stable are top-yield rankings? Heatmap cells show separate top-k means; pooled text summaries weight each valid observed-date/top-k comparison equally. Sample: date comparisons=7938.

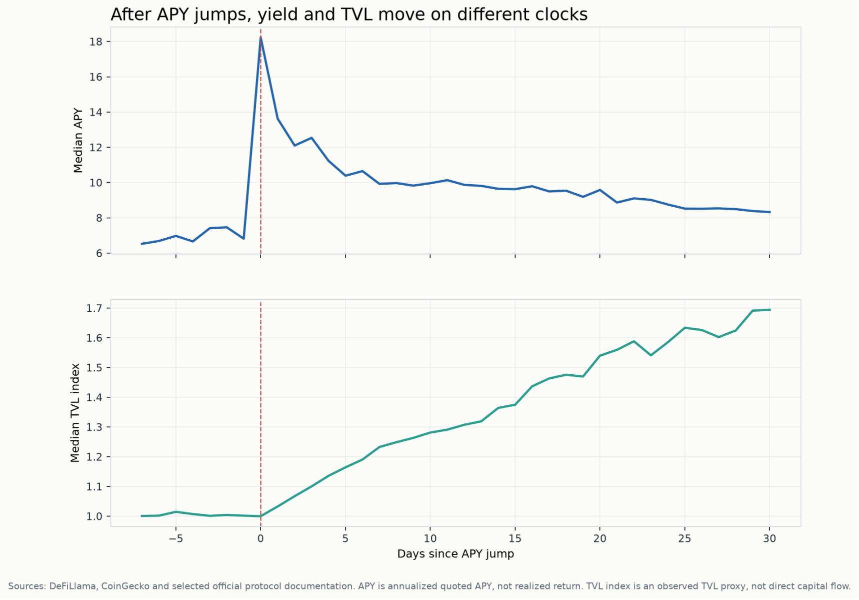
The top-yield set changes materially as the comparison horizon widens.

Comparison-weighted average churn rises from **13.5%** at one day to **33.3%** at thirty days.

Metric	Value
1 day	13.5%
7 days	23.6%
30 days	33.3%

Key reading
This is why the report avoids a best-pool leaderboard.

Do APY jumps and TVL move together?



Are APY jumps followed by TVL changes and compression? Event-time patterns are observational and use TVL change only as a proxy. Sample: event-time rows=38.

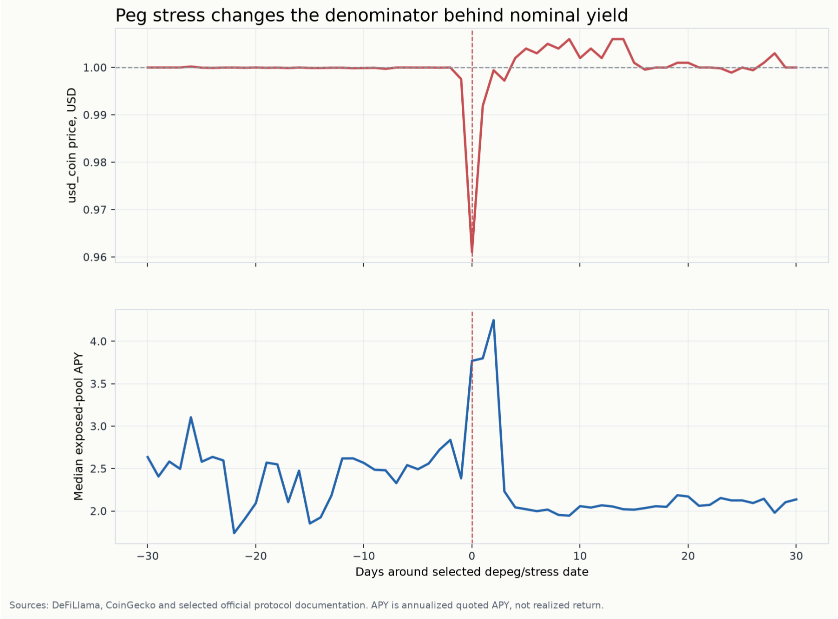
APY jumps and normalized TVL response are aligned in event time.

The pattern is observational: TVL can move because of deposits, withdrawals, price effects, accounting changes, migrations or source revisions.

Metric	Value
Events	453
Day -7	1.001
Day 30	1.694

Key reading
APY and TVL should be read on separate clocks.

What happens around peg stress?



How do APY and price behave around stablecoin peg stress? A depeg window makes nominal APY harder to interpret without price context. Sample: event-window rows=61.

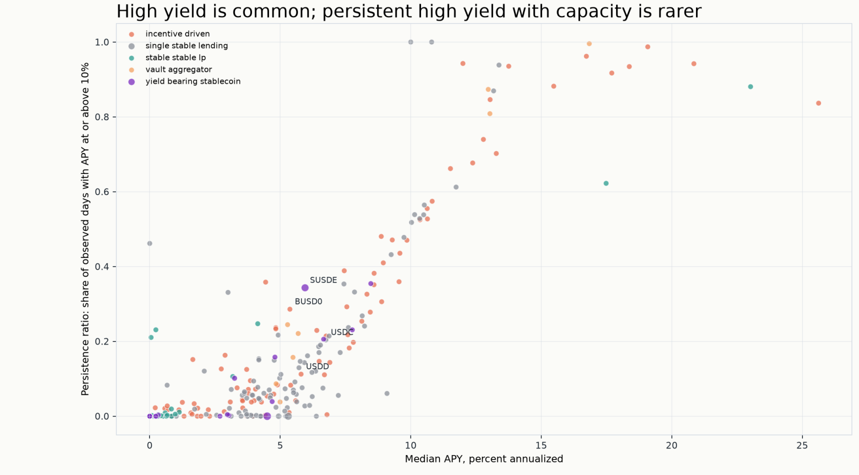
The selected event is **usd_coin** on **2023-03-12**.

The minimum observed price is **0.9611 USD**, so nominal APY needs stablecoin price context.

Metric	Value
Min price	0.9611
Peak dev.	0.0389
Pools	10

Key reading
Peg stress changes the denominator behind nominal yield.

How should yield be compared without ranking?



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

Which pools combine APY, persistence and TVL? Only a subset sits above median APY, persistence and TVL simultaneously; this is a joint threshold screen, not a recommendation. Sample: pools=250.

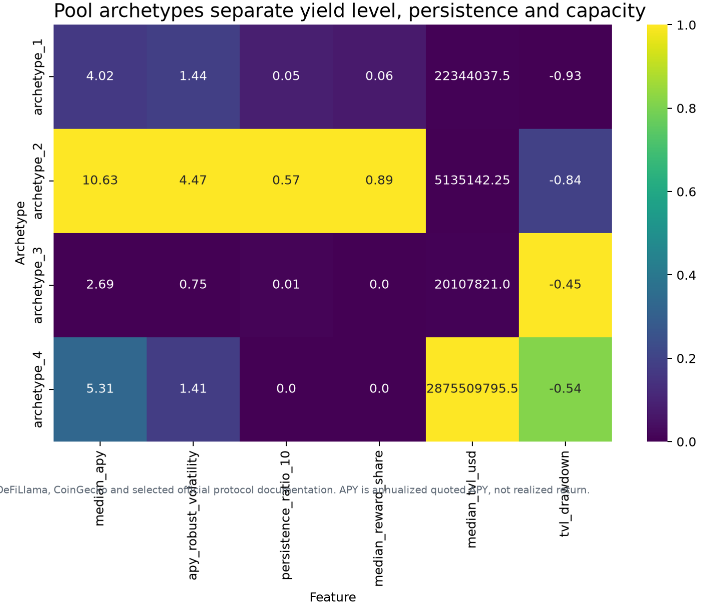
38 of **250** pools clear sample medians for APY, persistence and TVL simultaneously.

The joint screen keeps trade-offs visible instead of hiding them inside a score.

Metric	Value
Candidates	38
Share	15.2%
Pools	250

Key reading
The joint screen is descriptive, not financial advice.

Do pools cluster into interpretable profiles?



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

Do pools cluster into interpretable profiles? Exploratory clusters summarize patterns but are not labels of safety or quality. Sample: pools=250.

Archetypes summarize pool-level patterns across APY, persistence, TVL, reward share and volatility.

They are useful for discussion but are not risk labels, safety ratings or recommendations.

Metric	Value
archetype_1	138
archetype_3	61
archetype_2	48
archetype_4	3

Key reading
Clusters help describe heterogeneity; they do not replace the research questions.

Robustness checks

Check	Episodes	Median days	Censored
threshold_5	3,688	2.0	2.7%
threshold_10	2,668	2.0	1.2%
threshold_20	964	2.0	0.8%
min_tvl_500000	2,536	2.0	1.3%
min_tvl_1000000	2,470	2.0	1.3%
min_tvl_5000000	1,758	2.0	1.1%
min_history_90	2,668	2.0	1.2%
min_history_180	2,668	2.0	1.2%
min_history_365	1,919	2.0	0.7%
max_gap_0	2,683	2.0	1.2%
max_gap_1	2,668	2.0	1.2%
max_gap_3	2,646	2.0	1.2%
winsor_0.01_0.99	2,668	2.0	1.2%

Threshold sensitivity
Counts change, but median duration stays 2.0 days at 5, 10 and 20 percent APY.

Interpretation
Short duration is robust; it is not a universal forecast for every future episode.

Limitations and responsible interpretation

Concept	Boundary
APY	Quoted annualized APY is not realized return.
TVL	Observed TVL is not direct wallet-level capital flow.
Peg risk	Price deviations do not fully measure counterparty, legal, collateral or redemption risk.
Protocol risk	Mechanism labels are descriptive, not safety ratings.
Entity resolution	Bridged assets, renamed pools and migrations can remain ambiguous.
Visual ethics	The report avoids unsupported causality and recommendation framing.

No recommendation boundary
The project never identifies a best pool. Every pool-level label is descriptive evidence for the research question.

Reproducibility and deliverables

Task	Command
Install	uv sync --extra dev
Sample	uv run python scripts/reproduce_all.py --mode sample
Full	uv run python scripts/reproduce_all.py --mode full
Report	uv run python scripts/build_report.py --mode full

Deliverable	Path
Report PDF	outputs/report/stablecoin_yield_report.pdf
Report preview	outputs/report/rendered_preview/
Presentation	outputs/presentation/stablecoin_yield_presentation.pptx
Manifest	outputs/release_manifest.json

Final reading
High APY behaves like a regime, not a stable property. The joint screen keeps trade-offs visible without ranking pools.